HEAT TRANSFER IN NATURAL CONVECTION

Introduction: In contrast to the forced convection, Natural convection phenomenon is due to the temperature difference between the surface and the fluid and is not created by any external agency. The present experimental set up is designed and fabricated to study the natural convection phenomenon from a vertical cylinder in terms of the variation of local heat transfer coefficient and its comparison with the value obtained by using an appropriate correlation.

Apparatus: The apparatus consists of a cartridge heater with brass tube fitted in a rectangular duct in vertical position. The duct is open at the top and the bottom, and forms an enclosure and serves the purpose of undisturbed surrounding. One side of the duct is made up of Perspex of visualisation. An electric heating element is kept in the vertical tube, which inturn heats the tube surface. The heat is lost from the tube to the surrounding air by natural convection. The temperature of the vertical tube is measured by seven thermocouples. The heat input to the heater is measured by an ammeter and a voltmeter and is varied by a dimmerstat. The vertical cylinder with the thermocouple positions is shown in the fig. The tube surface is polished to minimize the radiation losses.

Specifications:

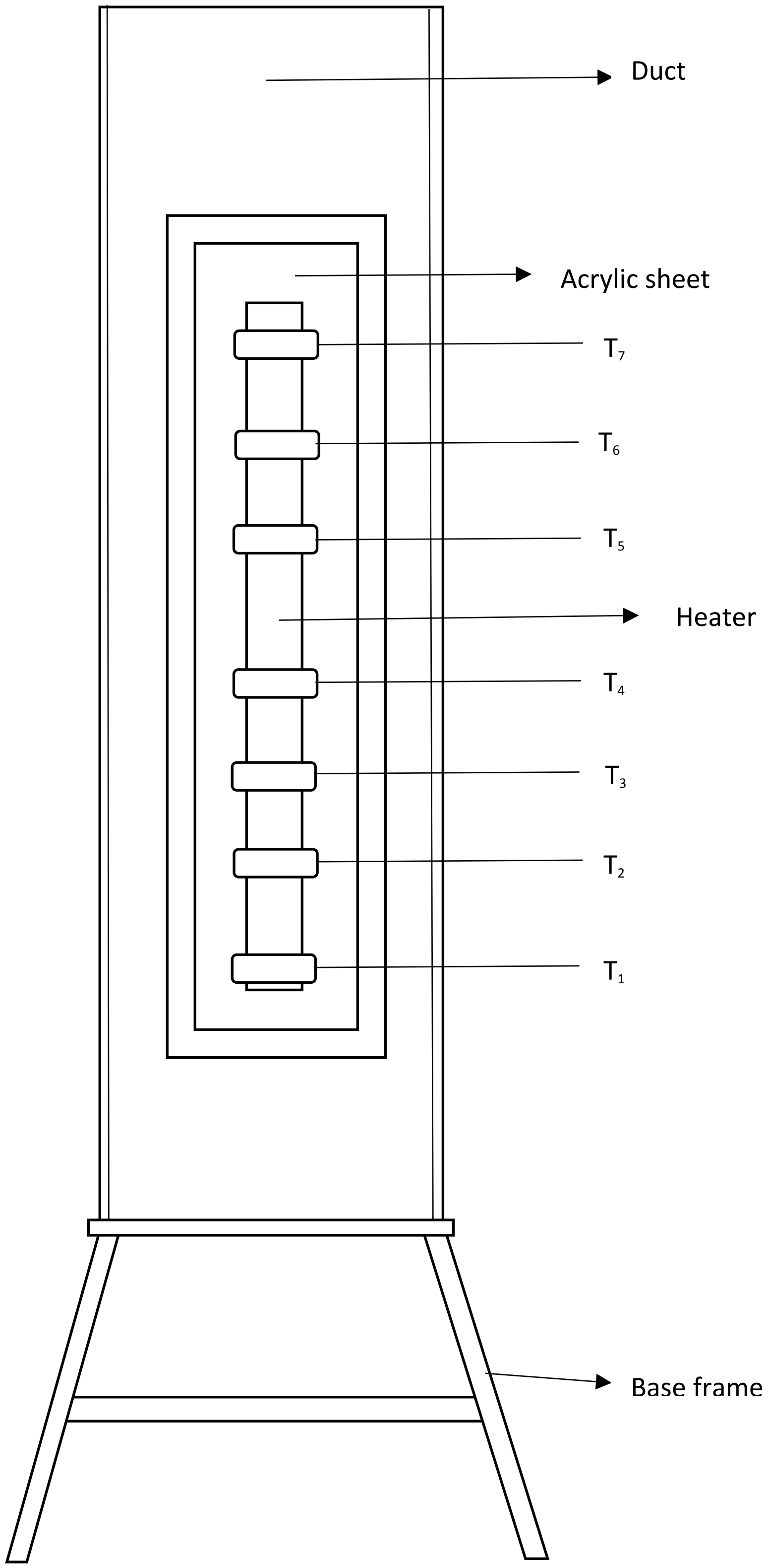
Diameter of the tube (d)	= 25mm
Length of the tube (L)	= 400mm
Duct size	= 210mm x 210mm x 750mm
No. of thermocouples	= 7 (K type)

Thermocouple no. 8 reads the temperature of the air in the duct.

Temperature indicator	= 0 to 300^{0}C, Multichannel type, calibrated for chromel alumel thermocouples
Ammeter	= 0 to 2 amp
Voltmeter	= 0 to 100/200V
Dimmerstat	= 2amp/ 230V
Heater	= cartridge type (400 watts)

Aim: To determine the local heat transfer coefficient for a vertical tube losing heat by natural convection.

Theory: When a hot body is kept in a still atmosphere, heat is transferred to the surrounding fluid by natural convection. The fluid layer in contact with the hot body gets heated, rises up due to the decrease in its density and the cold fluid rushes in from bottom side. The process is continuous and the heat transfer takes place due to the relative motion of the hot and fluid particles.



Experimental setup to determine heat transfer in natural convection

The heat transfer coefficient is given by:-

h = Average surface heat transfer coefficient (W/m^2 K).
q = Heat transfer rate (watts).
A_s = Area of the heat transferring surface (m^2)
T_a = Ambient temperature in the duct (K)
T_s = Average surface temperature (K)
T_s = $(T_1 + T_2 + T_3 + T_4 + T_5 + T_6 + T_7)/7$

The surface heat transfer coefficient, of a system transferring heat by natural convection depends on the shape, dimensions and orientation of the fluid and the temperature difference between the heat transferring surface and the fluid. The dependence of 'h' on all the above mentioned parameters is generally expressed in terms of non-dimensional groups as follows:

Where,
is called Grashof number
is called the Prandtl number
L is the characteristic dimension of surface
k is the thermal conductivity of the fluid
ν is the kinematic viscosity of the fluid
μ is the dynamic viscosity of the fluid
c_p is the specific heat at constant pressure of the fluid
β is the coefficient of volumetric expansion of the fluid
g is the acceleration due to gravity
$\Delta T = T_s - T_a$
A and n are constants depending upon the shape and orientation of the heat transferring surface.
For gas, K^{-1}
Where,

For a vertical cylinder losing heat by natural convection, the constants A and n have been determined and the following empirical correlations have been obtained.

$Nu = 0.56\ (Gr.Pr)^{0.25}$ for $10^4 < Gr.Pr < 10^8$
$Nu = 0.13\ (Gr.Pr)^{1/3}$ for $Gr.Pr > 10^8$
All the properties of the fluid are determined at the mean film temperature.

Observations:

1. Cylinder outer diameter = 25mm
2. Length of the cylinder = 400mm
3. Input to the heater = VI

Sl no.	Thermocouple readings (°C)								T_s (°C)	T_f (K)	Gr	Ra	h_{th} (W/m²K)	h_{exp} (W/m²K)
	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_a						
1														
2														
3														
4														
5														

Sl no:	Power given to heater		
	Voltage (volt)	Current (ampere)	Power (watts)
1			
2			
3			
4			
5			

Calculations:

Average surface temperature, T_s = $(T_1 + T_2 + T_3 + T_4 + T_5 + T_6 + T_7)/7$

Ambient temperature $T_a = T_8$ = ________________

Surface area of heat transfer, As

Average heat transfer coefficient, h_{av} =

Grashof number, Gr =

Where, ΔT = (Ts – Ta)

Prandtl number, Pr (from table) =

Kinematic viscosity of the fluid, =

Gr x Pr =

Nusselt number, Nu =

Procedure:

1. Put ON the supply and adjust the dimmerstat to obtain the required heat input.
2. Wait till the fairly steady state is reached, which is confirmed from temperature readings (T_1 to T_7).
3. Note down the surface temperature at various points.
4. Note the ambient temperature (T_8).
5. Repeat the experiment at various heat inputs.

Precautions:

1. Adjust the temperature indicator to ambient level by using compensation screw, before starting the experiment if needed.
2. Keep Dimmerstat to zero volt position and increase it slowly, don't exceed 80 volts.
3. Use the proper range of ammeter and voltmeter.
4. Operate the changeover switch of temperature indicator gently from 1 to 8 positions.

Results:

HEAT TRANSFER IN FORCED CONVECTION

Introduction: The apparatus consists of a blower unit fitted with the test pipe. The test section is surrounded by a Nichrome band heater. Four thermocouples are embedded on the test section and two thermocouples are placed in the air stream at the entrance and exit of the test section to measure the air temperature. Test pipe is connected to the delivery side of the blower along with the orifice to measure flow of air through the pipe. Input to the heater is given through a dimmerstat. It is to be noted that only a part of the total heat supplied is utilized in heating the air. A temperature indicator with cold junction compensation is provided to measure temperatures of pipe wall at various points in the test section. Airflow is measured with the help of orifice meter and the water manometer fitted on the board. Details of the test pipe are given in the figure.

Specifications:

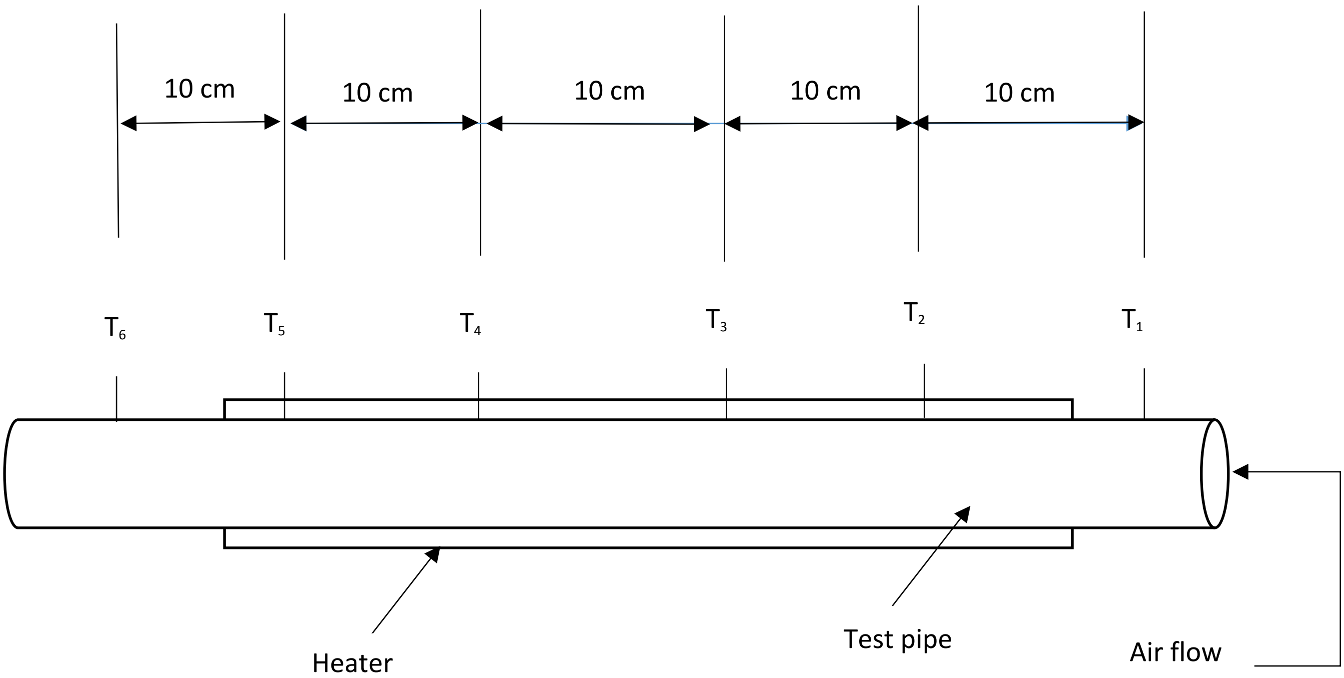
1.	Pipe diameter, Do	= 33mm
2.	Pipe diameter, Di	= 28mm
3.	Length of test section, L	= 400mm
4.	Blower ½ FHP motor	
5.	Orifice diameter, d	= 14mm, connected to water manometer
6.	Dimmerstat	= 0 to 2 amp, 260 volt, AC
7.	Temperature indicator	= range 0 – 300°C
8.	Voltmeter	= 0 – 100/200 volt
9.	Ammeter	= 0 – 2 amp
10.	Heater	= Nichrome wire heater wound on test pipe 400watt

Aim: To determine the heat transfer coefficient in forced convection of air in a tube and compare with the predicted value of heat transfer coefficient.

Theory: Forced convection in tubes can be either laminar or turbulent. Although laminar flows through ducts do not occur as extensively as turbulent flows, they do not occur in a number of situations in which size of the duct involved is very small or in which fluid involved has a high viscosity. Near the inlet to the duct or following the disturbance produced by a fitting such as a bend, there will be a region in which the characteristics of the flow and the heat transfer rates are changing relatively rapidly with the distance along the duct. Following this "entrance'' or "developing flow" region, the flow reaches a "fully developed" state in which basic characteristics of the flow are not changing with distance along the duct. In the fully developed flow region both the velocity profile and the temperature profile are not changing along the distance of the duct. However fully developed flow is attained in the flow well down stream of the entrance of the duct or bend or any other fitting in the duct or a region over which there is a change in the conditions at the duct wall. In the entrance region, the velocity and temperature fields are simultaneously developing; that is both are changing along the distance along the duct.

In some cases, there may be a long section of the duct where there is no heat transfer prior to section in duct where heat transfer is begin to take place. In such cases the velocity profile is essentially fully developed before heat transfer occurs and the flow is said to be "hydrodynamically developed" and "thermally developing" in the region of heat transfer.

Thermocouple positions of Forced Convection Apparatus



The Reynolds number for the flow in circular tube is defined as

Where,

v is the mean fluid velocity over the tube cross section.

D is the diameter of the duct

ρ is the density of the fluid

μ is the dynamic viscosity of the fluid

In a fully developed flow, critical Reynolds number corresponding to the onset of turbulence is $Re_{cr} = 2300$.

For laminar flow ($Re < 2300$) the hydrodynamic entry length may be written as

For laminar flow the thermal entry length may be written as

In contrast, for turbulent flow, conditions are nearly independent of Prandtl number, and to a first approximation, we shall assume = 10. The thermal conditions imposed on the surface of a tube are two types.

i. Uniform surface heat flux
ii. Uniform surface temperature

In a thermally fully developed flow of a fluid with constant properties, the local heat transfer coefficient, independent of x. Following empirical correlations are used in the analysis.

For Re < 2300, laminar flow

1. Pr >>1

Thermal entry length problem, unheated starting length

2. Re Pr > 10, 0.48<Pr<16700

Combined entry length problem

Observations:

1. Outer diameter of the pipe, Do = 33mm
2. Inner diameter of the pipe, Di = 28mm
3. Length of test section, L = 400mm
4. Orifice diameter, d = 14mm

Sl no.	Ammeter reading (A)	Voltmeter reading (V)	Temperature (^{o}C)						Manometer reading (m)
			T1	T2	T3	T4	T5	T6	

Calculations

Specific heat of air, Cp (from chart at Ta) = ___________________

Density of air, ρ_a = ___________________

Density of water, ρ_w = ____________________

Coefficient of discharge, Cd = ____________________

Gravitational acceleration, g = ____________________

Air discharge, Q =

= _____________________

Area of pipe, Ap =

= _____________________

Velocity of air at outlet = Q/Ap = ______________

Mass flow rate of air, m_a = Q . ρ_a . 3600

Heat carried away by air, q_a = m_a . Cp ($T_6 - T_1$)

= _______________________

Mean temperature, T_s = = __________

Mean temperature of air, T_a = (T_1 + T_6)/2 = __________

Heat transfer area, A = π D_i L = _______________

Heat transfer coefficient, h_a = =______________

For 2300 < Re < 10,000 thermally transition region

For Re > 10,000 Turbulent flow

, tube flow entrance region, 10 < L/D < 400

, fully developed turbulent flow, L/D ≥ 10,

0.6 ≤ Pr ≤ 16,700

Equations used:

1. The rate at which air is getting heated:

, in KJ/hr

Where,

m = mass flow rate of air (kg/hr)
Cp = specific heat of air (kJ/K-kg)
T = temperature rise in air (°C)

$$T = T_6 - T_1$$

$$M = Q.a$$

Where,

a	= density of air to be evaluated at $(T_1 + T_6)/2$ (kg/m^3)
Q	= volume flow rate (m^3/sec)

Where,

q_a	= the rate at which air is getting heated (watt)
A	= the test section area
	= π D_i L (in m^2)
T_a	= average temperature of the air $(T_1 + T_6)/2$
T_s	= average surface temperature
	=
Cd	= coefficient of discharge = 0.64
h	= difference of water level in the manometer, (m)
ρ_w	= density of water (1000 kg/m^3)
ρ_a	= density of air (1.03 kg/m^3)
d	= diameter of the orifice (0.014m)

Thermal conductivity, k (at Ta) = ________________________

Actual Nusselts number, Nu = = ______________

Kinematic viscosity, ν = ________________________

Reynolds number, Re = = _______________

Prandtl number, Pr = ________________________

Theoretical Nusselt number =

2. Reynolds number

Where,

V	= velocity of air
	=
ν	= kinematic viscosity (m^2/sec), to be evaluated at $(T_1 + T_6)/2$

3. Nusselt number

Where,

k	= thermal conductivity of air (W/m K), at $(T_1 + T_6)/2$

4. Prandtl number

Where,

C_p = specific heat of the fluid (kJ/K kg)

u = velocity of the fluid (m/sec)

k = thermal conductivity of the fluid (W/m K)

The appropriate correlation for turbulent flow through closed conduits is the Dittus – Boelter correlation.

Precautions:

1. Keep the dimmerstat at zero position before switching ON the power supply.
2. Start the blower unit.
3. Increase the voltmeter gradually.
4. Do not stop the blower in between the testing period.
5. Do not disturb thermocouples while testing.
6. Operate selector switch of the temperature indicator gently.
7. Do not exceed 200 watts.

Procedure:

1. Start the blower and adjust the flow by means of gate valve to some desired difference in the manometer level.
2. Start the heating of the test section with the help of dimmerstat and adjust the desired heat input with the help of voltmeter and ammeter.
3. Take readings of all the six thermocouples at an interval of 10 min until the steady state is reached.
4. Note down the heater input.

Result:

HEAT TRANSFER THROUGH COMPOSITE WALLS

Introduction: The apparatus consists of a central heater sandwiched between two sheets. Three types of slabs are provided on both sides of heater, which forms a composite structure. A small hand press frame is provided to ensure the perfect contact between the slabs. A dimmerstat is provided for varying the input to the heater and measurement of input is carried out by a voltmeter and ammeter.

Thermocouples are embedded between interfaces of the slabs, to read the temperature at the surface.

The experiments can be conducted at various values of input and calculation can be more accordingly.

Apparatus: The apparatus consists of a central heater sandwiched between two sheets. Three types of slabs are provided on both sides of heater, which forms a composite structure. A small hand press frame is provided to ensure the perfect contact between the slabs. A dimmerstat is provided for varying the input to the heater and measurement of input is carried out by voltmeter and ammeter.

Thermocouples are embedded between the slabs to read the temperature at the surface.

The experiments can be conducted at various values of input.

Specifications:

1. Slab assembly arranged symmetrically on both sides of the heater.
2. Heater : Nichrome heater wound on mice former and insulator with control unit capacity 300 watt.
3. Heater control unit : 0-230volt, 0-2ampere single phase Dimmerstat
4. Voltmeter : 0-100,200 volt
5. Ammeter : 0 -2 ampere
6. Temperature indicator: 0 – 200 °C (digital type),

Theory: By Fourier law of heat transfer,

For calculating effective area of heat transfer through composite wall, central half diameter area is only considered (heat is assumed to flow through central portion of large diameter plates).

Thermal conductivity of wall

Rate of heat supplied Q = 0.86 V I

For calculating the thermal conductivity of composite walls, it is assumed that due to large diameter of the plates, heat flowing through central portion is uni-directional i.e. axial flow.

Observations:

Composite slabs:

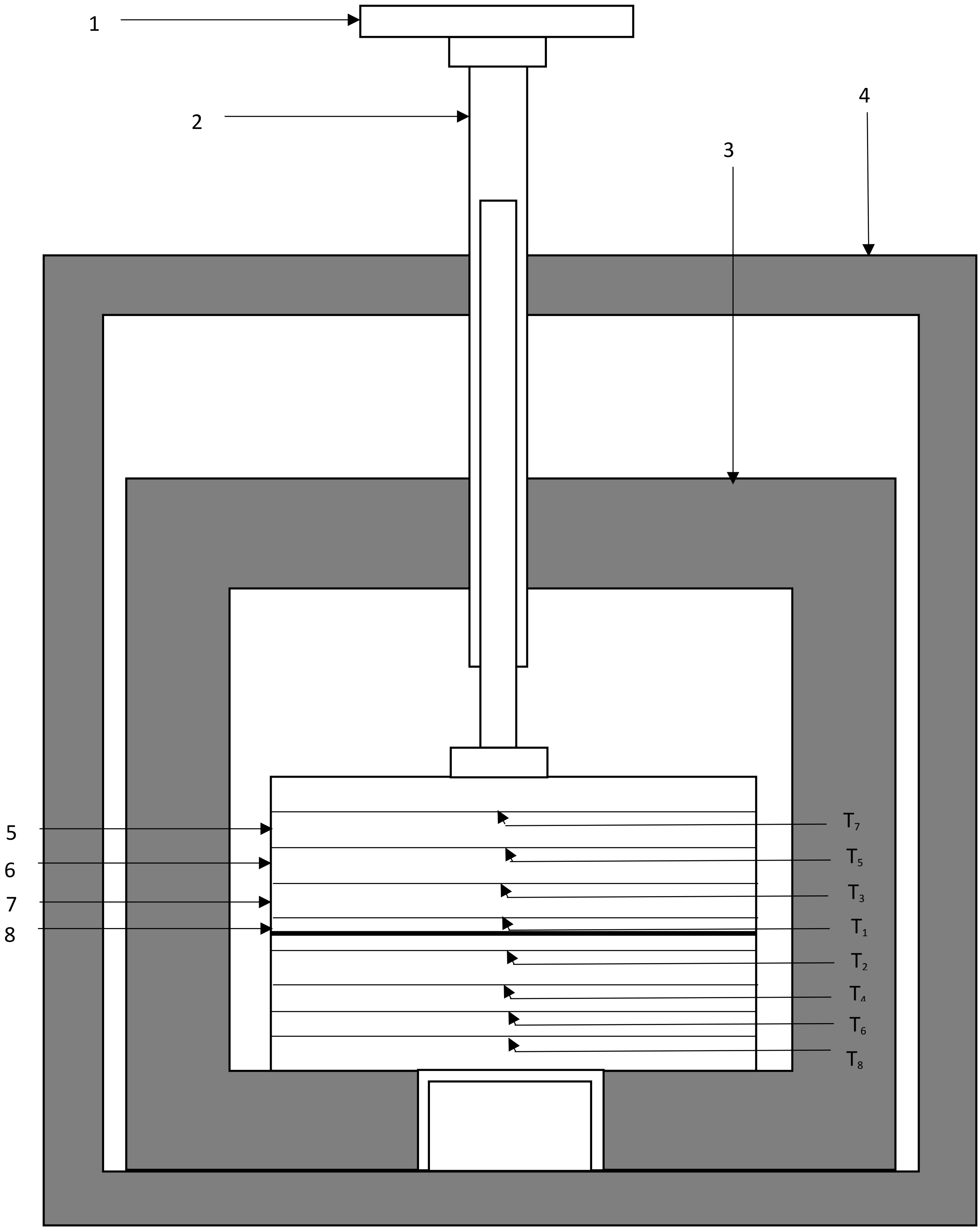
Wall thickness:

1. Mild steel = 1.2cm
2. Bakelite = 1.0cm
3. Plywood = 1.6cm

Slab diameter = 30cm

Sl no.	Set I	Set II	Set II
Voltmeter reading (V)			
Ammeter reading (A)			
Heat supplied (kcal/hr) Q = 0.86 x W			
THERMOCOUPLE READING (^oC)			
T_1			
T_2			
T_3			
T_4			
T_5			
T_6			
T_7			
T_8			

Mean reading:



1 – Hand Wheel

2 – Screw

3 – Cabinet

4 – Fabricated Frame

5 – Bakelite Plate

6 – Press Wood Plate

7 – C I Plate

8 – Heater

Thus for calculations, central half diametric area, where uni-directional flow is assumed, is considered. Accordingly thermocouples are fixed at close to centre of the plates.

Now,

Heat flux, kcal/hr-m^2

Where,

d is the half diameter of the plates

Total thermal resistance of composite slab, R_{total} =

Thermal conductivity of composite slab, K_{comp} =

Where,

b is the total thickness of the composite slab

Aim: To determine total thermal resistance and thermal conductivity of composite wall.

Procedure:

1. Arrange the plates in proper fashion symmetrical on both sides of the heater plates.
2. See that plates are symmetrically arranged on both sides of the heater plates.
3. Close the box by cover sheet to achieve steady environmental conditions.
4. Start the supply of heater by varying the dimmerstat, adjust the input at the desired value.
5. Take readings of all the thermocouples at an interval of 10 minutes until fairly steady temperatures are achieved and the rate of rise is negligible.
6. Note down the reading in observation table.

Precautions:

1. Keep dimmerstat to zero before start.
2. Increase the voltage slowly.
3. Keep all the assembly undisturbed.
4. Remove air gap between plates by moving hand-press gently.
5. While removing the plates do not disturb the thermocouples.
6. Operate selector switch of temperature indicator gently.

Result:

EMISSIVITY MEASUREMENT APPARATUS

Introduction:

All substances at all temperature emit thermal radiation. Thermal radiation is an electromagnetic wave and does not require any material medium for propagation. All bodies can emit radiation and have also the capacity to absorb all or a part of the radiation coming from the surrounding towards it.

An idealized black surface is one, which absorbs all the incident radiation with reflectivity and transmissivity equal to zero. The radiant energy per unit time per unit area from the surface of the body is called, as the emissivity of the surface is the ratio of the emissive power of the surface to the emissive power of a black surface at the same temperature. It is noted by E.

For black body absorptivity = 1 and by the knowledge of Kirchoff's Law emissivity of the black body becomes unity.

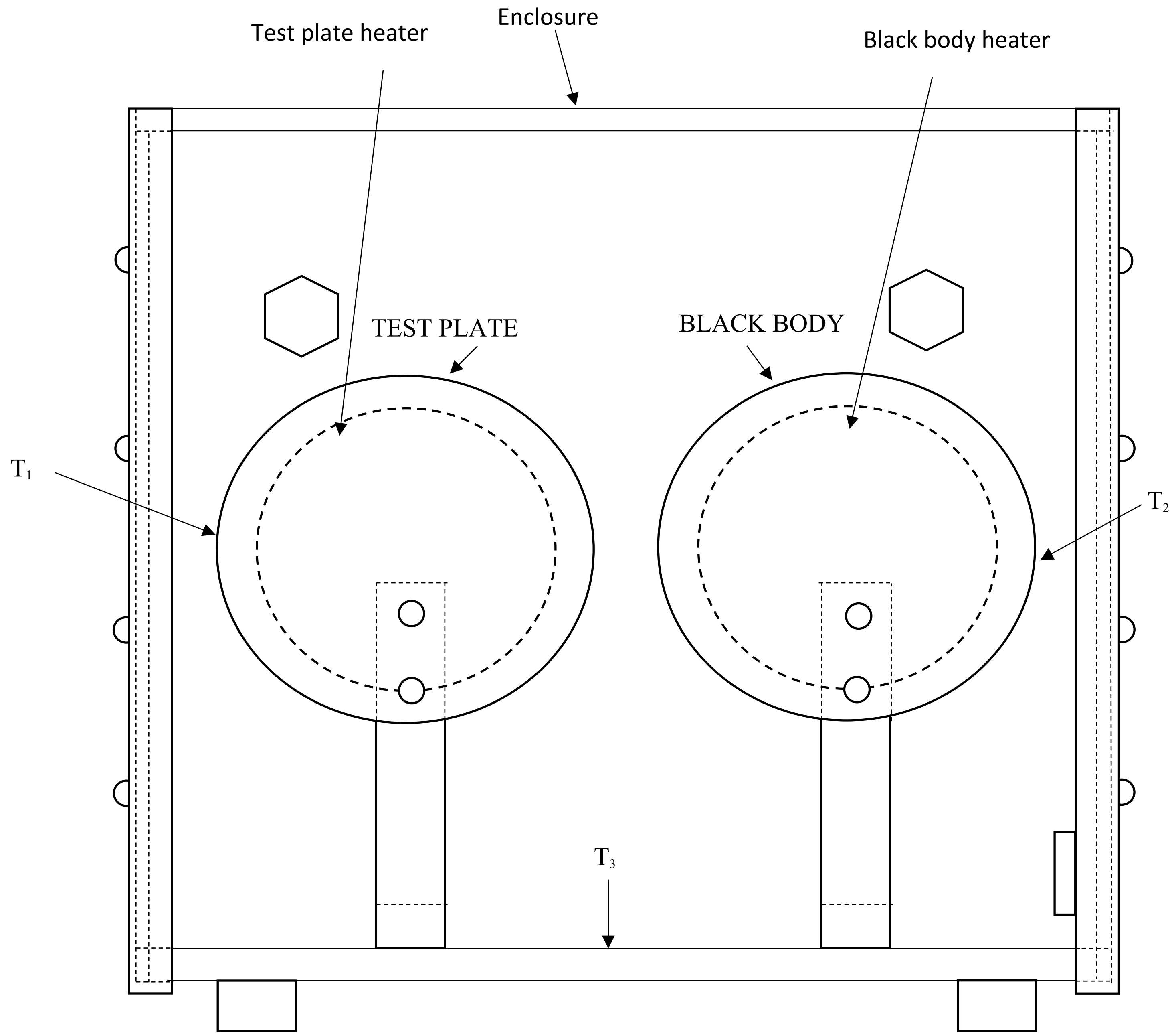
Emissivity being a property of the surface depends on the nature of the surface and temperature.

It is obvious from the Stefan Boltzmann's Law that the prediction of emissive power of a surface requires knowledge about the values of its emissivity and therefore much experimental research in radiation has been concentrated on measuring the values of emissivity as function of surface temperature. The present experimental set up is designed and fabricated to measure the property of emissivity of the test plate surface at various temperatures.

Table 1 gives approximate values of emissivity for some common materials for reference.

TABLE – 1

	MATERIAL	TEMPERATURE	EMISSIVITY
Metals	Polished copper, Steel, Stainless Steel, Nickel Alluminium (Oxidised)	20^0C 90 - 540^0C	0.15 increases with temperatures 0.20 to 0.33
Non- Metals	Brick, Wood, Marble, water	20 – 100^0C	0.80 to 1



SCHEMATIC OF EMISSIVITY MEASUREMENT APPARATUS

Under steady state conditions:

Heater input black plate, W_1 =

Heater input to test plate, W_2 =

Area of plates, A

Where,

d = Diameter of plate (in m)

By using the Stefan Boltzmann Law:

$(W_1 - W_2) = (E_b - E)\ \sigma\ A\ (T_s^4 - T_d^4)/0.86$ (In MKS units)

$(W_1 - W_2) = (E_b - E)\ \sigma\ A\ (T_s^4 - T_d^4)$ (In SI units)

Where,

T_b = Temperature of black plate (in °K)

T_a = Ambient temperature (in °K)

E_b = Emissivity of black plate (To be assumed equal to unity.)

E = Emissivity of non-black test plate

σ = Stefan Boltzmann constant.

σ = 4.876×10^{-8} Kcal/ hr-m^2- K^4 (In MKS units)

σ = 5.67×10^{-8} W/m^2 K^4 (In SI units)

Specifications:

1. Test Plate = Ø 165mm
2. Black Plate = Ø 165mm

$\Big\}$ Material: Aluminium

3. Heater for (1) Nichrome strip wound on mica sheet sandwiched between two mica sheets.
4. Heater for (2) as above capacity of heater = 200 watts each approx.
5. Dimmerstat for (1) = 0 – 2A, 0 – 260V
6. Dimmerstat for (2) = 0 – 2A, 0 – 260V
7. Voltmeter = 0 – 100 – 200V
8. Ammeter = 0 – 2 Amp.
9. Enclosure size 580mm x 300mm x 300mm approximately with one side of perpex sheet.
10. Thermocouples = Choromel Alumel – (3Nos).
11. Temperature indicator= 0 – 300^{0}C.
12. D. P. D. T. Switch.

Observations:

Sr. No.	BLACK PLATE			TEST PLATE			ENCLOSURE TEMP.
	V1	I1	T_b	V_2	I_2	T_s	T_a ^0C

Calculations:

$$(W_b - W_s) = (E_b - E_s)\ \sigma\ A\ (T_s^4 - T_b^4)$$

$$Q_b = \sigma A\ E_b\ (T_s^4 - T_D^4)$$

$$Q_b = \sigma E\ A\ (T_s^4 - T_D^4)$$

Where,

Q_b = heat input to disc coated with lamp black (in W).

W_b = V_1 I_1 Watts

Q_s = heat input to Specimen disc. (in W)

W_s = V_2 I_2

σ = Stefan Boltzmann Constant = 4.876×10^{-8} Kcal/hr m^2 °K^4

σ = 5.67×10^{-8} W/M^2 K^4

E = Emissivity of specimen to be determined (absorption)

This fact could be verified by performing the experiments at various values of T_s and E can be plotted in a graph.

Aim: To determine the emissivity of a given grey body.

Procedure:

1. Gradually increase the input to the heater to black plate and adjust it to some value viz. 30, 50, 75 watts and adjust the heater input to test plate slightly less than the black plate 27, 35, 55 watts etc.
2. Check the temperature of the two plates with small time intervals and adjust the input of test plate only, by the dimmerstat so that the two plates will be maintained at the same temperature.
3. This will required some trial and error and one has to wait sufficiently (more than one hour or so) to obtain the steady state condition.
4. After attaining the steady state condition record the temperatures. Voltmeter and Ammeter readings for both the plates.

5. The same procedure is repeated for various surface temperature in increasing order.

Precaution:

1. Use stabilized A. C. single-phase supply (preferably).
2. Always keep the dimmerstats at zero position before start.
3. Use the proper voltage range on voltmeter.
4. Gradually increase the heater inputs.
5. See that the black plate is having a layer of lamp black uniformly.

There is a possibility of getting absurd results if the supply voltage is fluctuating or if the input is not adjusted till the satisfactory steady state condition reached.

Result:

STEFAN BOLTZMANN APPARATUS

Introduction:

The most commonly used law of thermal radiation is the Stefan Boltzmann Law which states that thermal radiation heat flux or emissive power of a black surface is proportional to the fourth power of absolute temperature of the surface and is given by:

= eb = σT^4 (kcal / hr.m^2. k^4) in MKS and (W/m^{2}k^4) in SI units
The constant of proportionality σ is called the Stefan Boltzmann Constant and has the value is 4.876 x 10^{-8} (kcal / hr.m^2. k^4) and SI unit K^4 (5.67 x 10^{-8} W/m^{2}k^4)

The Stefan Boltzmann law can be derive by integrating the planks law over the entire spectrum of wave length, through historically it is worth nothing that the Stefan Boltzmann law was independently developed before planks law.

The object of this experimental set up is to measure the value of this constant fairly close, by an easy arrangement.

The radiation energy falling on D from enclosure is given by:

$$E = A_D\,(T)^4\,\sigma \text{ ....(1)}$$

Where A_D = Area of the disc – D in m^2

The emissivity of the disc D is assumed to be unity (Black disc). The radiant energy disc D is emitting into the enclosure will be:

$$E_1 = A_D\,(T)_5^4\,\sigma \text{ ....(2)}$$

Net heat in put to disc D per unit time is given by (1) – (2):

$$E - E1 = \sigma\, A_D (T^4 - T_5^4) \text{..................(3)}$$

If the disc D has mass m and specific heat S then a short time after D is inserted in A.

$$m.\ s.\ (dT / dt)\ t \qquad = \sigma\, A_D (T^4 - T_5^4)$$

Observation:

1. Mass of the test disc (m) = 0.005Kg.
2. Specific heat of disc material (s) = 0.1 Kcal / Kg °C (Copper disc) – MKS Unit
   = 0.41868 KJ / Kg °C – SI Unit

Thermocouple No.	Temperature °C
T_1	
T_2	
T_3	
T_4	

Average temperature in °C =

=

3. Temperature of disc D at the instant when it is inserted (T_5)

   T_5 (in °C) =

   T_5 (in °K) =

4. Temperature time response of the disc.

Use the Disc thermocouple on T. I. and note down T_5 at the time interval of 5 sec.

Time (Sec.) t	Temperature (T_5) °C
5	
10	
15	
20	

Plot the graph of dT against t

5. Obtain slope from the graph

   (dT / dt) at t = 0 = ____________________°C /sec

   = ____________________°C/hr

6. Value of σ can be obtain by using

   kcal/ hr K^4 m^2 (M K S)

   W /m^2 K^4 (S I Units)

Kcal/ hr K^4 m^2 (M K S)

W /m^2 K^4 (S I Units)

In this equation (dT / dt) t = 0 denotes the rate of rise temperature of the disc D at the instant when its temperature is T5 and will vary with T5. It is clearly best measured at time t = 0 before heat conducted from A to D begins to have any significant effect.

This is obtained from plot of temperature rise of D with respect to time and obtaining its slope at t = 0 when Temperature = T_5. This will be the required value of dT / dt at t = 0. The thermocouple mounted on disc is to be used for this purpose.

Note that the disc D with its insulating sleeve S, is placed quickly in position and start the timer and record the temperature at fixed time intervals. The whole process is completed in about 30 seconds of time.

Longer D is left in position; the greater is the probability of errors due to heat conduction from A to D.

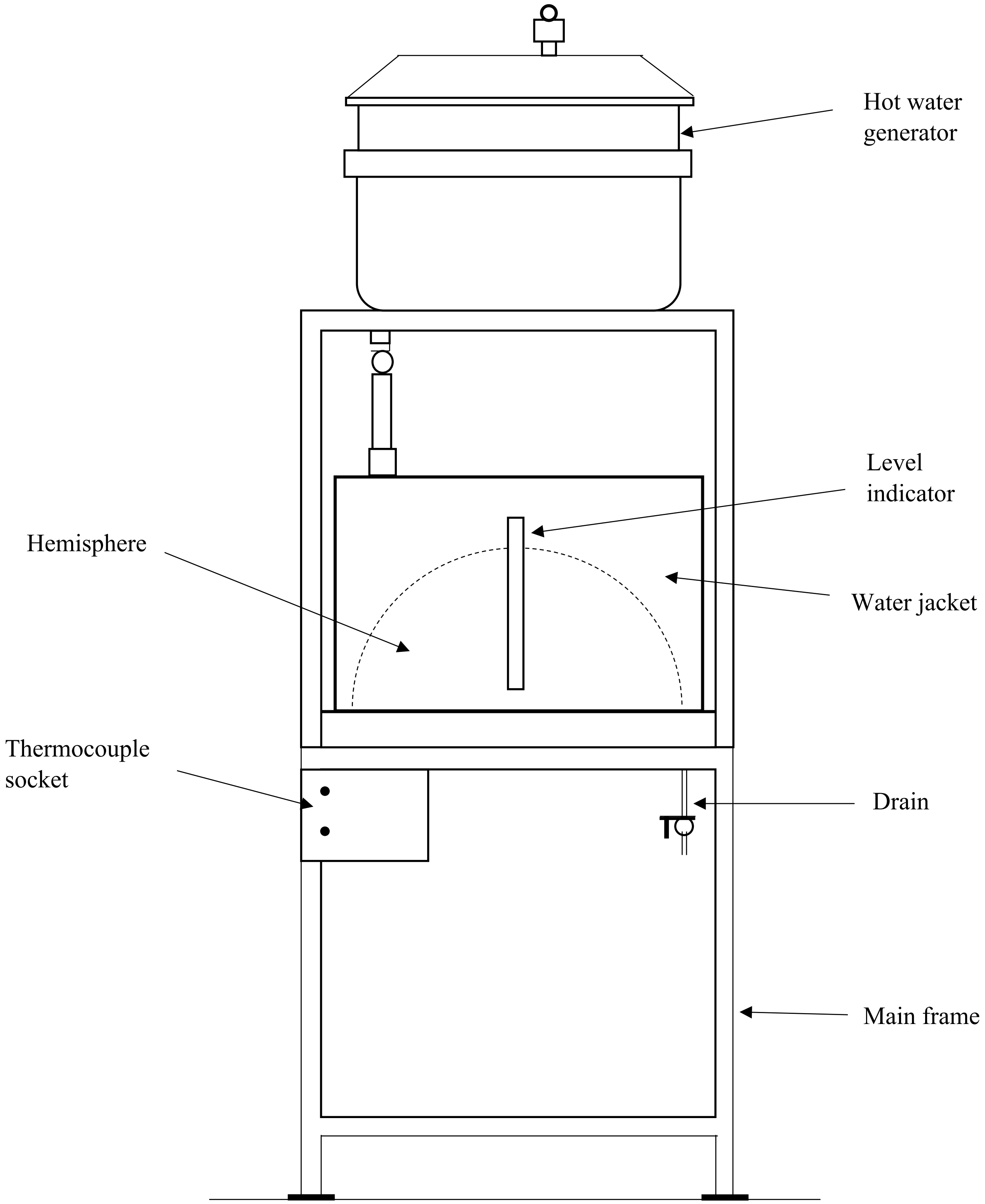
Apparatus:

The apparatus, as illustrated in is centred on a flanged copper hemisphere B fixed on a flat non-conducting plate A. The outer surface of B is enclosed in a metal water jacket used to heat B to some suitable constant temperature. The hemisphere B is chosen solely on the grounds that it simplifies the task of water between B & C

Four chromel alumel thermocouple are attached to various points on surface of B to measure its mean temperature.

The disk D, which is mounted in an insulating bakelite sleeve S is fitted in hole drilled in the centered of based plate A. the of S is conveniently supported for underside of A. A chromel alumel thermocouple is used to measure the temperature of D (T_5). Thermocouple is mounted on the disk to study the rise of its temperature.

When the disk is inserted at the temperature T_5 ($T_5 \geq T$ ie the temperature of the enclosure), the response of the temperature change of this with time is used to calculate the Stefan Boltzmann constant.



Schematic Layout of Stefan Boltzmann Apparatus

Specifications:

1. Hemispherical enclosure = 200mm.
2. Suitable sized water jacket for hemisphere.
3. Base plate, bakelite diameter = 240mm.
4. Sleeve size, diameter = 44mm
5. Fixing arrangement for sleeve
6. Test disc diameter = 25mm
7. Mass of test disc = 0.004Kg.
8. Specific heat, S of the test disc = 0.1 kcal/Kg ^{0}C

(Ref. Mech. Engg. Hand book for copper 0.41868Kj/Kg ^{0}C In SI unit)

9. No. of thermocouples mounted on B = 4
10. No. of thermocouples mounted on D = 1
11. Temperature indicator digital 0.1^{0}C least count, 0-200^{0}C with built in cold junction compensation and a timer set for 5 Sec. To display temperature rise of the disc.
12. Immersion water heater of suitable capacity = 3KW
13. Tank for hot water.

The surface of B and A forming the enclosure are blackened by using lamp black to make their absorptivities to be approximately unity. The copper surface of the disc D is also blackened.

Aim: To determine the Stefan Boltzmann constant.

Experimental procedure:

1. The water in the tank is heated by immersion heater up to a temperature of about 90^{0}C.
2. The disc D is removed before pouring the hot water in the jacket.
3. The hot water is poured in the water jacket.
4. The hemispherical enclosure B and A will come to some uniform temperature T in short time after filling the hot water in jacket. The thermal inertial of hot water is quite adequate to presence significant cooling in the time required to conduct the experiment.
5. The enclosure will soon come to thermal equilibrium conditions.
6. The disc D is now inserted in A at a time when its temperature is say T5 (to be sensed by separate thermocouple).
7. The water in the tank is heated by immersion heater up to a temperature of about 90^{0}C.
8. The disc D is removed before pouring the hot water in the jacket.
9. The hot water is poured in the water jacket.
10. The hemispherical enclosure B and A will come to some uniform temperature T in short time after filling the hot water in jacket. The thermal inertial of hot water is

quite adequate to presence significant cooling in the time required to conduct the experiment.

11. The enclosure will soon come to thermal equilibrium conditions.
12. The experiment is repeated for obtaining better result.

Precaution:

1. Use stabilized A. C. single-phase supply (preferably).
2. Always keep the dimmerstats at zero position before start.
3. Use the proper voltage range on VOLTMETER.
4. Gradually increase the heater inputs.
5. See that the black plate is having a layer of lamp black uniformly.
6. There is a possibility of getting absurd results if the supply voltage is fluctuating or if the input is not adjusted till the satisfactory steady state condition reached.

Result:

SHELL AND TUBE HEAT EXCHANGER

Introduction:

Shell and tube heat exchanger finds application in a variety of industries and is, without doubt, one of the most widely used exchangers. It has a series of tubes which is enclosed by a shell. One fluid flows inside the tubes while the other liquid flows over the outside walls of the tubes which, basically, is the shell. It's highly recommended for places where there's a need for high heat transfer coefficient as the number of tubes can be increased depending on the need. Due to its unique shape, it finds use in high pressure applications. The present experimental setup is designed and fabricated to study the heat transfer in shell and tube heat exchanger and to find its effectiveness, LMTD and overall heat transfer coefficient at various temperatures and flow rates of hot and cold fluid.

Description of Apparatus:

The experimental setup consists of 18 tubes of shell, 200 mm outside diameter at each side and length 1m long. The effective length of the tube is 1020 mm and the outside diameter is 12.5 mm. There are two tube passes. The shell is made of mild steel. The flow rate of water through the tubes is measured using rota meters. Thermocouples are provided for measuring the inlet and outlet temperatures of hot and cold waters

Specification:

Outside diameter of the tubes, d_o	=	12.5 mm
Effective length of the tubes, l	=	1020 mm
Total number of tubes, n	=	18 (9 on each sides)
Number of passes on the tube side	=	2

THERMOCOUPLE DETAILS

T_1	=	cold water inlet
T_2	=	cold water outlet
T_3	=	hot water inlet
T_4	=	hot water outlet

Observations:

Hot fluid Inlet Temperature T_{hi}	Hot fluid outlet Temperature T_{ho}	Cold fluid Inlet temperature T_{ci}	Cold fluid Outlet Temperature T_{co}	Mass flow rate of hot fluid M_h	Mass flow rate of cold fluid M_c	LMTD	Heat transfer Q	Overall heat transfer coefficient U	Effectiveness ε

Calculation:

1. Heat transfer from hot water

   $Q_h \quad = m_h\, c_{ph}\, (T_{hi} - T_{ho})$

   =

2. Heat gained by cold fluid

   $Qc \quad = m_c\, c_{pc}\, (T_{co} - T_{ci})$

   =

3. Amount of heat transfer occurred

   $Q \quad = (Q_h + Q_c) / 2$

   =

4. Logarithmic Mean Temperature Difference LMTD θ_{lm}

   =

5. Capacity ratio

   =

Equations used:

1. Heat transfer from hot water

$Q_h = m_h\ c_{ph}\ (T_{hi} - T_{ho})$

m_h = Mass flow rate of hot water in m^3/sec

measurement is taken in cc/sec should be convert it into m^3/sec

C_{ph} = Specific heat of hot water (J/kg K)

2. Heat gained by cold fluid

$Q_c = m_c\ c_{pc}\ (T_{co} - T_{ci})$

m_c = Mass flow rate of cold water in m^3/sec

C_{pc} = Specific heat of cold water (J/kg K)

3. Amount of heat transfer occurred

$Q = (Q_h + Q_c) / 2$

4. Logarithmic Mean Temperature Difference LMTD θ_m

where θ_m= log mean temperature difference

$\theta_1 = T_{hi} - T_{co}$

$\theta_2 = T_{ho} - T_{ci}$

5. Capacity ratio

6. Temperature ratio

7. Total outer surface of the tubes

d_o	=	Outer diameter of tube
n	=	no of tubes

l = Length of the pipe

8. Correction factor F

Correction factor F for Shell and tube heat exchanger is determined from the Correction factor plot (available in data book), corresponding to the values of capacity ratio and temperature ratio.

The Correction factor plot for heat exchanger with one shell pass and two, four or multiple tube passes is as shown below.

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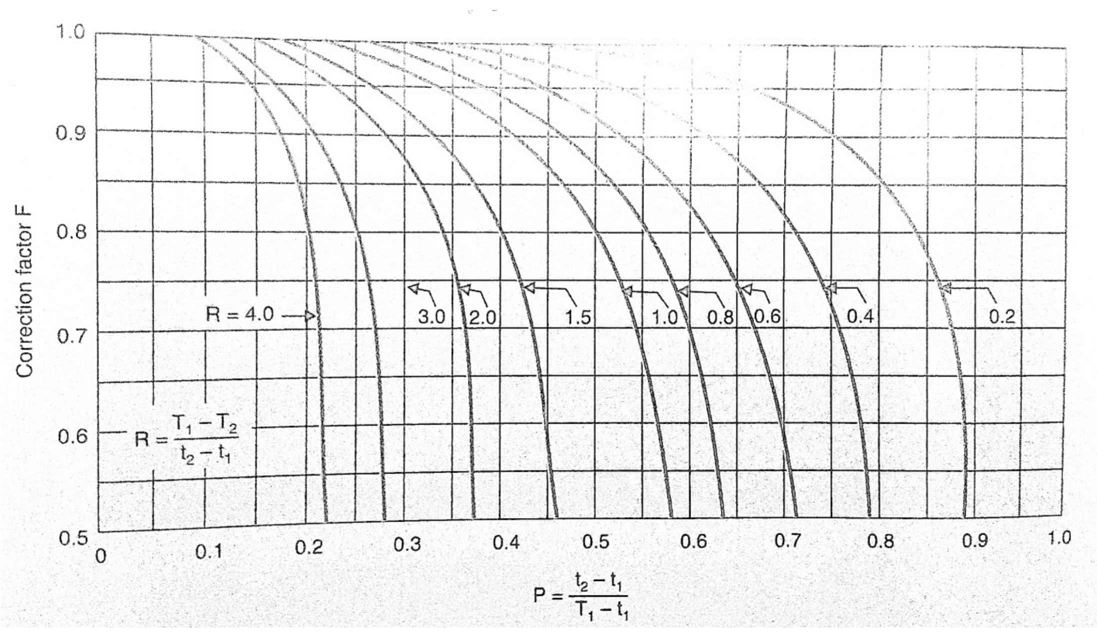

9. Overall heat transfer coefficient based on outer surface area of inner tube

$$U_o = Q / (A_o \times LMTD \times F)$$

=

$$A_o = \pi\, d\, l\, n$$

=

10. Q_{max} = C_{min} x (T_{hi} - T_{ci}) =

11. Overall heat transfer coefficient based on outer surface area of inner tube

$U_o = Q / (A_o \times LMTD \times F)$ W/m²K

Where $A_o = \pi\, d\, l$ m^2

d = outer diameter of tube

l = length of the tube

F = correction factor for HE with 1 shell pass and 2 tube pass

12. Effectiveness

Effectiveness, $\varepsilon = Q / Q_{max}$

$\varepsilon = (T_{hi} - T_{ho}) / (T_{hi} - T_{ci})$ if $m_h c_h < m_c c_c$

$\varepsilon = (T_{co} - T_{ci}) / (Thi - T_{ci})$ if $m_c c_c < m_h c_h$

Where m_h and m_c are mass flow rate of hot and cold fluid respectively

c_h and c_c are specific heat of hot and cold fluids respectively.

Aim: To find the effectiveness, LMTD and overall heat transfer coefficient of the shell and tube heat exchanger.

Procedure:

1. Allow the water to flow from the sump tank or tap to the pump
2. Switch on the heaters to produce hot water
3. Allow the hot water to circulate through the tube by operating the corresponding valves.
4. Allow the cold water to flow through the shell side of the heat exchanger
5. Allow sufficient time so that the unit attains a steady state
6. Note down the temperature reading and water flow rates using the rotameter

Result:

PARALLEL FLOW HEAT EXCHANGER

Introduction:

In a parallel flow heat exchangers, both the fluids, hot fluid and cold fluid, enters from the same end and travel in same direction. The flow arrangement and the variation of the temperature in the heat exchanger is shown below. It is evident from the graph that the temperature difference between hot and cold fluids goes on decreasing from inlet to outlet. Therefore this kind of heat exchanger needs more area for heat transfer.

The present experimental setup is designed and fabricated to study the heat transfer in shell and tube heat exchanger and to find its effectiveness, LMTD and overall heat transfer coefficient at various temperatures and flow rates of hot and cold fluid.

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Description of the Apparatus:

The apparatus consist of a concentric tube heat exchanger. The hot fluid used is hot water and the cold fluid used is cold water. The hot water is obtained from the Geyser of heat capacity 3kW and the hot fluid pass through the inner tubes in the heat exchanger. Cold fluid flows over the tube in which the hot fluid flows and the cold water can be admitted from anyone of the end of the heat exchanger enabling the heat exchanger to run as a parallel or counter flow heat exchanger. This can be adjusted by operating the different valves provided. However for parallel flow heat exchanger we enable the hot and the cold fluid to enter from the same end.

A measuring jar and a stopwatch is used to measure the flow rate of hot and cold fluid. The temperature of the fluid can be measured by using thermocouples with digital display indicator the outer tube is provided with insulation to minimize the heat loss to the surroundings.

OBSERVATION TABLE (for parallel flow)

SI NO	Hot fluid inlet temp T_{hi}	Hot fluid outlet temp T_{ho}	Cold fluid inlet temp T_{ci}	Cold fluid outlet temp T_{co}	Log Mean Temp Diff. LMTD	Flow rate of hot water m_h (Kg/s)	Flow rate of cold water m_c (Kg/s)	Amount of heat transfer occurred Q (W)	Overall heat transfer coefficient U	Effectiveness €
1										
2										
3										
4										

Calculations (for parallel flow)

1. Heat transfer from hot water

   $Q_h = m_h C_{ph} (T_{hi} - T_{ho})$

   m_h = Mass flow rate of hot water=

   C_{ph} = Specific heat of hot water (J/kg K) =

2. Heat gained by cold fluid

   $Qc = m_c C_{pc} (T_{co} – T_{ci})$

   m_c = Mass flow rate of cold water =

   C_{pc} = Specific heat of cold water (J/kg K) =

3. Amount of heat transfer occurred

   $Q = (Q_h + Q_c) / 2$

4. LMTD $\theta_m = (\theta_1 – \theta_2)/ \ln (\theta_1/\theta_2)$

   $\theta_1 = T_{hi} – T_{ci}$ (for parallel flow HE) =

   $\theta_2 = T_{ho} - T_{co}$ (for parallel flow HE) =

Specifications:

Specimen material	:	Copper tube
Size of the specimen	:	12.5mm x 1500mm long
Outer shell material	:	G I
Size of the outer shell	:	40mm dia
Geyser capacity	:	1 litre, 3 kW

Equations used:

1. Heat transfer from hot water

   $Q_h = m_h\ C_{ph}\ (T_{hi} - T_{ho})$

   m_h = Mass flow rate of hot water

   C_{ph} = Specific heat of hot water (J/kg K)

2. Heat gained by cold fluid

   $Qc = m_c\ C_{pc}\ (T_{co} – T_{ci})$

   m_c = Mass flow rate of cold water

   C_{pc} = Specific heat of cold water (J/kg K)

3. Amount of heat transfer occurred

   $Q = (Q_h + Q_c) / 2$

4. LMTD $\theta_m = (\theta_1 – \theta_2)/ \ln (\theta_1/\theta_2)$

   where θ_m= log mean temperature difference

   $\theta_1 = T_{hi} – T_{ci}$ (for parallel flow HE)

   $\theta_2 = T_{ho} - T_{co}$ (for parallel flow HE)

   $\theta_1 = T_{hi} – T_{co}$ (for counter flow HE)

   $\theta_2 = T_{ho} - T_{ci}$ (for counter flow HE)

5. Overall heat transfer coefficient based on outer surface area of inner tube

   $U_o = Q / (A_o \times LMTD)$ W/m^{2}K

   Where $A_o = \pi\ d\ l$ m^2

   d = outer diameter of tube = 12.5mm

l = length of the tube = 1.5m

$A = \pi \times 0.0125 \times 1.5\ m^2 = 0.0589\ m^2$

6. Effectiveness

Effectiveness € = Q / Q_{max}

$m_h c_h$ =

$m_c c_c$ =

C_{min} =

$Q_{max} = C_{min} \times (T_{hi} - T_{ci})$

€ = Q / Q_{max}

7. Overall heat transfer coefficient based on outer surface area of inner tube

$U_o = Q / (A_o \times LMTD)$ W/m^{2}K

Where $A_o = \pi\, d\, l$ m^2

d = outer diameter of tube

l = length of the tube

8. Effectiveness

Effectiveness, $\varepsilon = Q / Q_{max}$

$\varepsilon = (T_{hi} - T_{ho}) / (T_{hi} - T_{ci})$ if $m_h c_h < m_c c_c$

$\varepsilon = (T_{co} - T_{ci}) / (T_{hi} - T_{ci})$ if $m_c c_c < m_h c_h$

Where m_h and m_c are mass flow rate of hot and cold fluid respectively

c_h and c_c are specific heat of hot and cold fluids respectively.

Aim: To find the effectiveness, LMTD and overall heat transfer coefficient of the parallel flow heat exchanger.

Procedure:

1. First switch on the unit panel
2. Start the flow of cold water through the annulus and run the heat exchanger as parallel flow heat exchanger
3. Switch on the geyser provided on the panel and allow the flow through the inner tube by regulating the valve
4. Adjust the flow rate of hot water and cold water by using the valve arrangement in the apparatus
5. Allow the apparatus to run undisturbed i.e. keep the flow rate of cold and hot fluid same until steady state condition is obtained.
6. Note down the flow rate of hot and cold fluids and the temperature on hot and cold water sides(at inlet and outlet)
7. Now adjust the valves to get different flow rates and so as to take different set of readings.

Result:

COUNTER FLOW HEAT EXCHANGER

Introduction:

In counter flow heat exchanger, the two fluids flow in opposite directions. The hot and the cold fluid enters in opposite ends. The flow arrangement and the temperature is shown below, the temperature difference between the two fluids remains more or less same. Due to this, counter flow heat exchangers provide maximum amount of heat transfer for given area.

The present experimental setup is designed and fabricated to study the heat transfer in shell and tube heat exchanger and to find its effectiveness, LMTD and overall heat transfer coefficient at various temperatures and flow rates of hot and cold fluid.

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Description of Apparatus:

The apparatus consist of a concentric tube heat exchanger. The hot fluid used is hot water and the cold fluid used is cold water. The hot water is obtained from the Geyser of heat capacity 3kW and the hot fluid pass through the inner tubes in the heat exchanger. Cold fluid flows over the tube in which the hot fluid flows and the cold water can be admitted from anyone of the end of the heat exchanger enabling the heat exchanger to run as a parallel or counter flow heat exchanger. This can be adjusted by operating the different valves provided. However for parallel flow heat exchanger we enable the hot and the cold fluid to enter from the same end.

A measuring jar and a stopwatch is used to measure the flow rate of hot and cold fluid. The temperature of the fluid can be measured by using thermocouples with digital display indicator the outer tube is provided with insulation to minimize the heat loss to the surroundings.

Observations (for counter flow)

SI NO	Hot fluid inlet temp T_{hi}	Hot fluid outlet temp T_{ho}	Cold fluid inlet temp T_{ci}	Cold fluid outlet temp T_{co}	Log Mean Temp Diff. LMTD	Flow rate of hot water m_h (Kg/s)	Flow rate of cold water m_c (Kg/s)	Amount of heat transfer occurred Q (W)	Overall heat transfer coefficient U	Effectiveness €
1										
2										
3										
4										

Calculations (for counter flow)

1. Heat transfer from hot water

Q_h $= m_h C_{ph} (T_{hi} - T_{ho})$

m_h = Mass flow rate of hot water=

C_{ph} = Specific heat of hot water (J/kg K) =

5. Heat gained by cold fluid

Q_c $= m_c C_{pc} (T_{co} – T_{ci})$

m_c = Mass flow rate of cold water =

C_{pc} = Specific heat of cold water (J/kg K) =

6. Amount of heat transfer occurred

$Q = (Q_h + Q_c) / 2$

7. LMTD $\theta_m = (\theta_1 – \theta_2)/ \ln (\theta_1/\theta_2)$

$\theta_1 = T_{hi} – T_{co}$ (for counter flow HE) =

$\theta_2 = T_{ho} - T_{ci}$ (for counter flow HE) =

Specifications:

Specimen material	:	Copper tube
Size of the specimen	:	12.5mm x 1500mm long
Outer shell material	:	G I
Size of the outer shell	:	40mm dia
Geyser capacity	:	1 litre, 3 kW

Equations used:

1. Heat transfer from hot water

   $Q_h = m_h \, C_{ph} \, (T_{hi} - T_{ho})$

   m_h = Mass flow rate of hot water

   C_{ph} = Specific heat of hot water (J/kg K)

2. Heat gained by cold fluid

   $Qc = m_c \, C_{pc} \, (T_{co} - T_{ci})$

   m_c = Mass flow rate of cold water

   C_{pc} = Specific heat of cold water (J/kg K)

3. Amount of heat transfer occurred

   $Q = (Q_h + Q_c) / 2$

4.

5. LMTD $\theta_m = (\theta_1 - \theta_2) / \ln(\theta_1/\theta_2)$

   where θ_m = log mean temperature difference

   $\theta_1 = T_{hi} - T_{ci}$ (for parallel flow HE)

   $\theta_2 = T_{ho} - T_{co}$ (for parallel flow HE)

   $\theta_1 = T_{hi} - T_{co}$ (for counter flow HE)

   $\theta_2 = T_{ho} - T_{ci}$ (for counter flow HE)

6. Overall heat transfer coefficient based on outer surface area of inner tube

   $U_o = Q / (A_o \times LMTD)$

   Where $A_o = \pi\, d\, l$

   d = outer diameter of tube = 12.5mm

   l = length of the tube = 1.5m

   $A = \pi \times 0.0125 \times 1.5 \; m^2 = 0.0589 \; m^2$

7. Overall heat transfer coefficient based on outer surface area of inner tube

   $U_o = Q / (A_o \times LMTD)$ $\quad W/m^2K$

   Where $A_o = \pi\, d\, l$ $\quad m^2$

   d = outer diameter of tube

   l = length of the tube

8. Effectiveness

   $m_h c_h$ =

   $m_c c_c$ =

   Effectiveness, $\varepsilon = Q / Q_{max}$

   $Q_{max} = C_{min} \times (T_{hi} - T_{ci})$

   $\varepsilon = (T_{hi} - T_{ho}) / (T_{hi} - T_{ci})$ $\quad$ if $m_h c_h < m_c c_c$

   $\varepsilon = (T_{co} - T_{ci}) / (T_{hi} - T_{ci})$ $\quad$ if $m_c c_c < m_h c_h$

   Where m_h and m_c are mass flow rate of hot and cold fluid respectively

   c_h and c_c are specific heat of hot and cold fluids respectively.

Aim: To find the effectiveness, LMTD and overall heat transfer coefficient of the counter flow heat exchanger.

Procedure:

1. First switch on the unit panel
2. Start the flow of cold water through the annulus and run the heat exchanger as counter flow heat exchanger

3. Switch on the geyser provided on the panel and allow the flow through the inner tube by regulating the valve
4. Adjust the flow rate of hot water and cold water by using the valve arrangement in the apparatus
5. Allow the apparatus to run undisturbed i.e. keep the flow rate of cold and hot fluid same until steady state condition is obtained.
6. Note down the flow rate of hot and cold fluids and the temperature on hot and cold water sides(at inlet and outlet)
7. Now adjust the valves to get different flow rates and so as to take different set of readings.

Result: